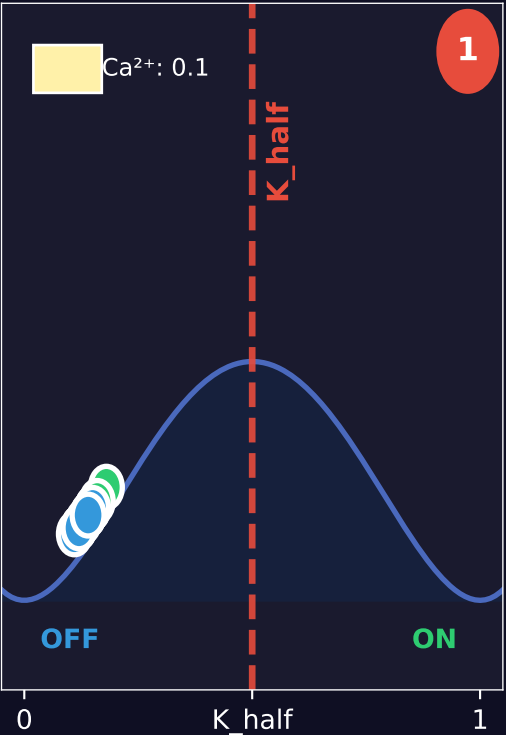
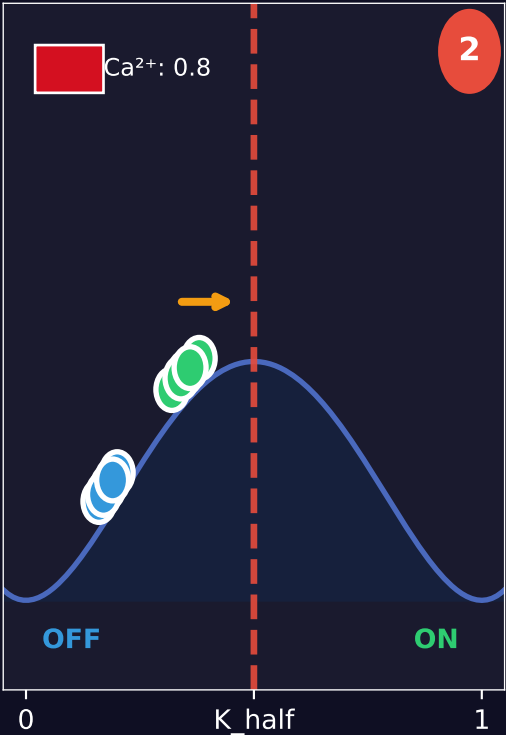


Ca²⁺ "Drags and Drops" CaMKII into Bistable States: A Mechanism for Bioelectric Pattern Memory

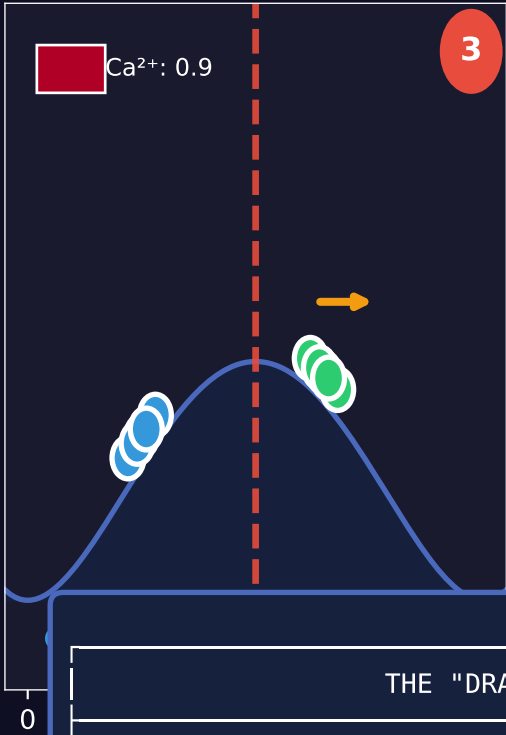
Phase 1: Initial State
All cells start in OFF state



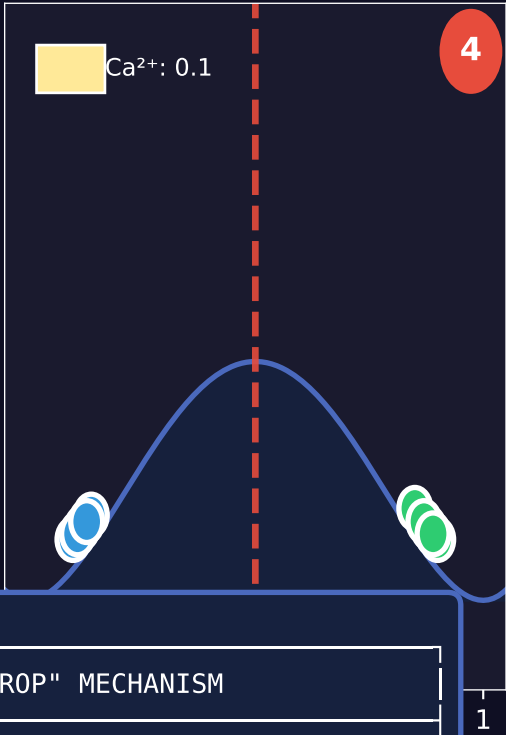
Phase 2: Ca²⁺ Rising
Vmem activates Ca²⁺ channels



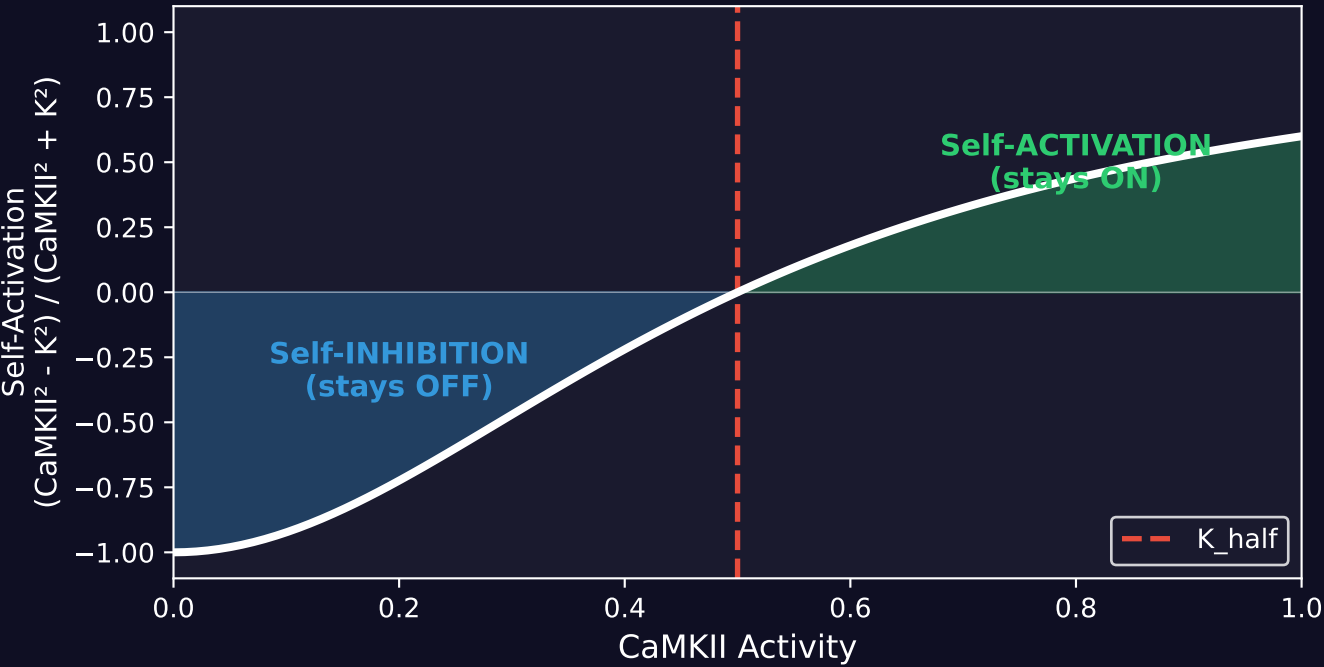
Phase 3: Ca²⁺ "Drops" Cells
Feature cells cross K_half



Phase 4: Bistable Memory
Pattern persists without Ca²⁺

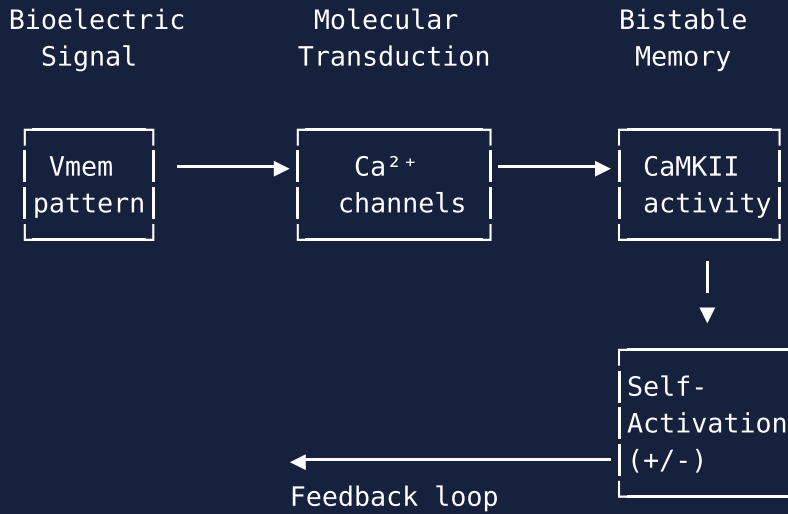


Competitive Dynamics: The Bistability Engine



● Feature cells (high Vmem → high Ca²⁺) ● Background cells (low Vmem → low Ca²⁺) - - - K_half: Decision boundary

THE "DRAG AND DROP" MECHANISM



KEY INSIGHT:

- Ca²⁺ "drags" CaMKII across K_half threshold
- Once past K_half: self-activation takes over (+)
- Below K_half: self-inhibition keeps it OFF (-)
- Pattern persists even after Ca²⁺ decays!

BIOPHYSICAL RELEVANCE:
• Converts transient bioelectric signals → stable memory
• Enables developmental pattern maintenance
• Similar to synaptic LTP/LTD mechanisms